

Re-imagining how national NUE is calculated and interpreted

Serra, J., Quemada, M., Haefele, S., Dobermann, A., Poole, C. and Lammell, J.

KEY CLAIM

National cropland NUE is most informative when crop mix effects are separated from management effects and countries are benchmarked against countries with similar cropping systems.

BRIEF JUSTIFICATION, REMAINING CHALLENGES

Nitrogen use efficiency (NUE) condenses national nitrogen budgets into a useful indicator, but it is not crop-agnostic. Crops differ in harvested nitrogen removal, residue removal and input requirements, so national NUE partly reflects the crops occupying cropland. Across Europe during 2000-2021, changes in crop mix altered national NUE trajectories substantially in several countries and changed the inferred direction of progress in ten. In some countries, the reported improvements disappeared when crop mix was held at its 2000 baseline; in others, the reported declines masked improvements in yield-input performance. National rankings also changed when countries were compared only with others sharing similar crop mixes.

A practical response is to retain conventional NUE but add two interpretive checks: a fixed crop mix trajectory that separates changes in yield-input performance from crop reallocation, and benchmarking within broadly similar cropping systems. These checks do not replace nitrogen surplus or other environmental indicators (e.g., nitrate leaching).

The main practical limitation is that national nitrogen budgets rarely allocate inputs by crop, so the crop mix contribution is diagnostic rather than a fully causal decomposition.

The communication challenge is equally important: this should not be presented as a new “correct” NUE, but as a transparent way to distinguish how crops are produced from what crops are produced.

POTENTIAL FOR SYNERGY WITH OTHER CONTRIBUTIONS

This contribution complements work on national nitrogen budgets, nitrogen footprints and surplus indicators by addressing how NUE should be interpreted when crop mix changes. Nitrogen footprints identify demand-side pressures, budgets quantify national nitrogen flows, and crop-mix-adjusted NUE helps distinguish changes in yield-input performance from changes associated with what is being produced.

QUESTIONS TO SEED THE DISCUSSION

1. When crop mix changes, what should constitute good practice for interpreting national NUE trends: a fixed crop mix trajectory, benchmarking against comparable cropping systems, or both?
2. How can crop mix adjusted NUE be combined with nitrogen surplus, environmental loss indicators and nitrogen footprints without making policy reporting too complex to use?

REFERENCES

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